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AI for Bharat Hackathon

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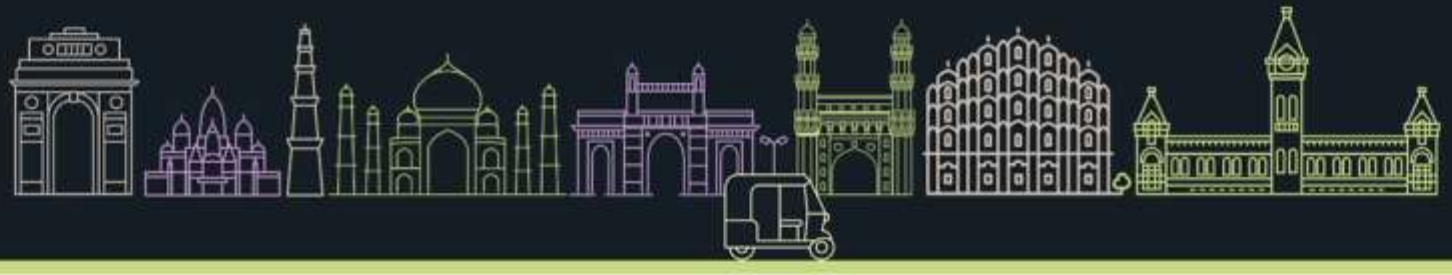


Team Name : TeamMAGNUS

Team Leader Name : SURAJ OJHA

Problem Statement : Small business owners struggle with manual financial record-keeping due to low digital literacy and language barriers. Traditional ledger apps require complex data entry, preventing adoption among shopkeepers who need fast, voice-based transaction logging in their native language.

Brief about the Idea:

BolKhata revolutionizes financial record-keeping for small businesses through voice-first technology powered by Sarvam AI. The core idea is to eliminate digital barriers by allowing shopkeepers to manage their entire ledger using natural speech in Hindi, English, or Hinglish.

The system works through intelligent voice recognition and natural language understanding. When a shopkeeper speaks a transaction like "Ramesh ne 500 rupaye diye" or "Sold goods to Priya for 1200 on credit," the app automatically identifies the customer name, amount, and transaction type without any manual input. It distinguishes between three transaction types: cash sales, credit sales, and payment receipts.

Beyond simple logging, BolKhata maintains comprehensive customer profiles with running balances, transaction history, and credit tracking. The dashboard provides real-time business insights including total sales, outstanding credits, and daily revenue trends. Each customer gets a digital passbook that can be shared via WhatsApp for transparency and payment reminders.

The application is built on enterprise-grade AWS infrastructure with EC2 for compute, RDS PostgreSQL for data persistence, and secured with HTTPS through Let's Encrypt. It's accessible at https://bolkhata.com, making it a production-ready solution that combines the simplicity of voice interaction with the power of cloud computing to serve India's 63 million MSMEs who need accessible financial tools.

Your solution should be able to explain the following:

1.Why AI is required in your solution?

AI is absolutely essential because it solves the fundamental barrier preventing small business adoption - the inability to use traditional keyboard-based interfaces. Your target users (shopkeepers, small business owners) have low digital literacy and prefer speaking in their native language over typing. AI enables three critical capabilities:-

First, Sarvam AI's speech-to-text engine converts natural spoken Hindi, English, or Hinglish into text, handling the acoustic variations and accents of Indian languages. Second, the Natural Language Understanding layer extracts structured transaction data (customer name, amount, transaction type) from unstructured conversational speech like "Ramesh ne 500 rupaye diye" without requiring users to fill forms or navigate menus. Third, the translation service bridges language barriers by converting regional language input into English for entity extraction, then presenting results back in the user's preferred language.

Without AI, users would need to manually type customer names, select transaction types from dropdowns, and enter amounts - creating friction that prevents adoption. AI transforms a 5-step manual process into a single voice command, making financial record-keeping accessible to non-tech-savvy users.

2.How AWS services are used within your architecture?

Your application uses a comprehensive AWS infrastructure deployed in the us-east-1 region:

EC2 t3.medium instances host both the Spring Boot banking service (port 8080) and FastAPI voice service (port 8000), providing 2 vCPUs and 4GB RAM for concurrent request handling. RDS PostgreSQL manages all transactional data with automated backups, storing users, customers, transactions, and audio file metadata with ACID compliance. Route 53 handles DNS management for bolkhata.com domain with A records pointing to your Elastic IP. VPC provides network isolation with public subnets for EC2 and private subnets for RDS, secured by security groups controlling inbound traffic on ports 22, 80, 443, 8080, and 8000.

Nginx acts as the reverse proxy, routing frontend requests to static files, /api requests to the banking service, and /voice requests to the voice service. Let's Encrypt SSL certificates provide HTTPS encryption with automatic renewal. IAM roles grant EC2 instances permissions to access RDS, S3, and CloudWatch without embedding credentials. The audio files are stored locally on EC2 in organized directories by user and date, with paths stored in PostgreSQL for retrieval.
This architecture provides production-grade scalability, security, and reliability with automated backups, SSL encryption, and proper network segmentation.

3.What value the AI layer adds to the user experience?

The AI layer transforms BolKhata from just another ledger app into a truly accessible solution for India's small business ecosystem:

Speed and convenience - users record transactions in under 3 seconds by speaking naturally instead of spending 30+ seconds navigating forms and typing. A shopkeeper can say "Priya ko 1200 ka maal udhar diya" while serving the next customer, without interrupting their workflow.

Zero learning curve - no training needed on how to use the app. Users speak the same way they would write in a paper ledger: "Ramesh gave 500 rupees" or "Sold to Suresh for 800 cash." The AI understands intent regardless of phrasing variations.

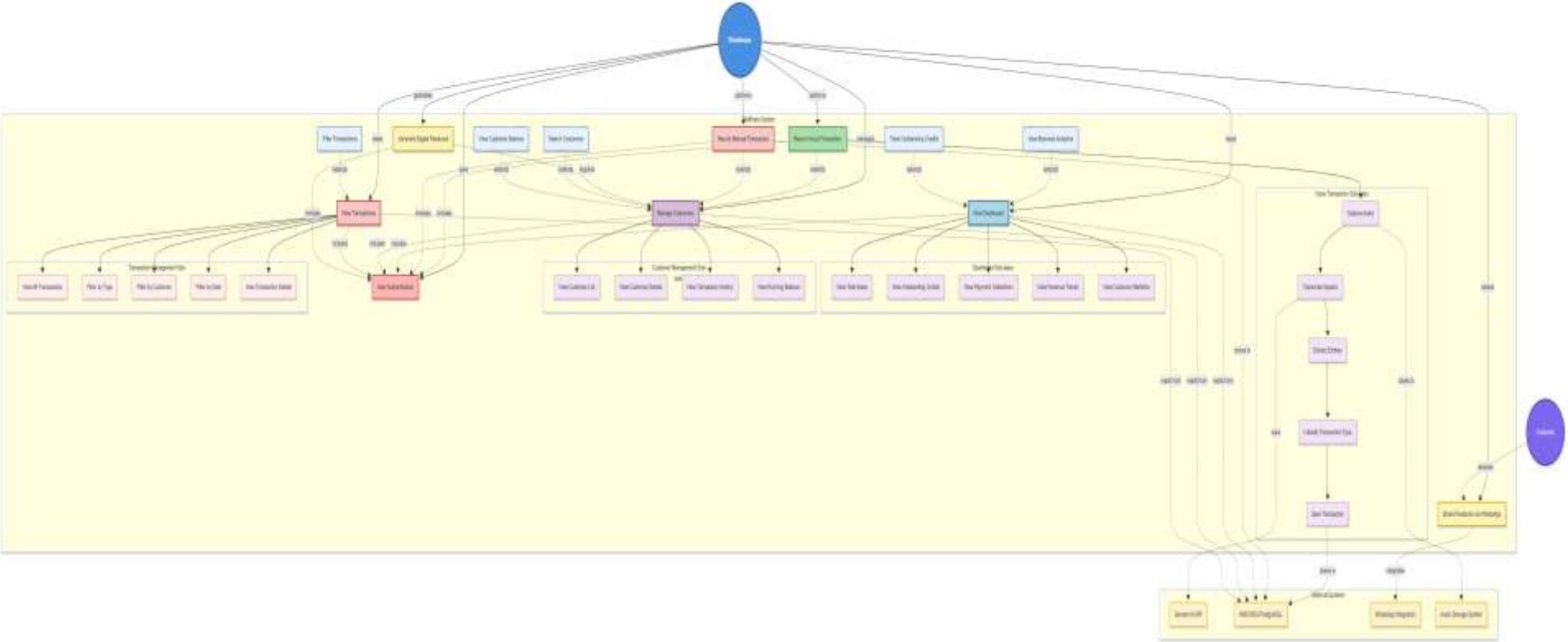
Language inclusivity - supports Hindi, English, and Hinglish with automatic translation, making it accessible to users who can't type in English but can speak their native language fluently. This removes the digital divide that excludes 85% of Indian MSMEs from using existing solutions.

Intelligent transaction classification - automatically distinguishes between cash sales (SALE_PAID), credit sales (SALE_CREDIT), and payment receipts (PAYMENT_RECEIVED) based on conversational context, eliminating the cognitive load of understanding accounting terminology.

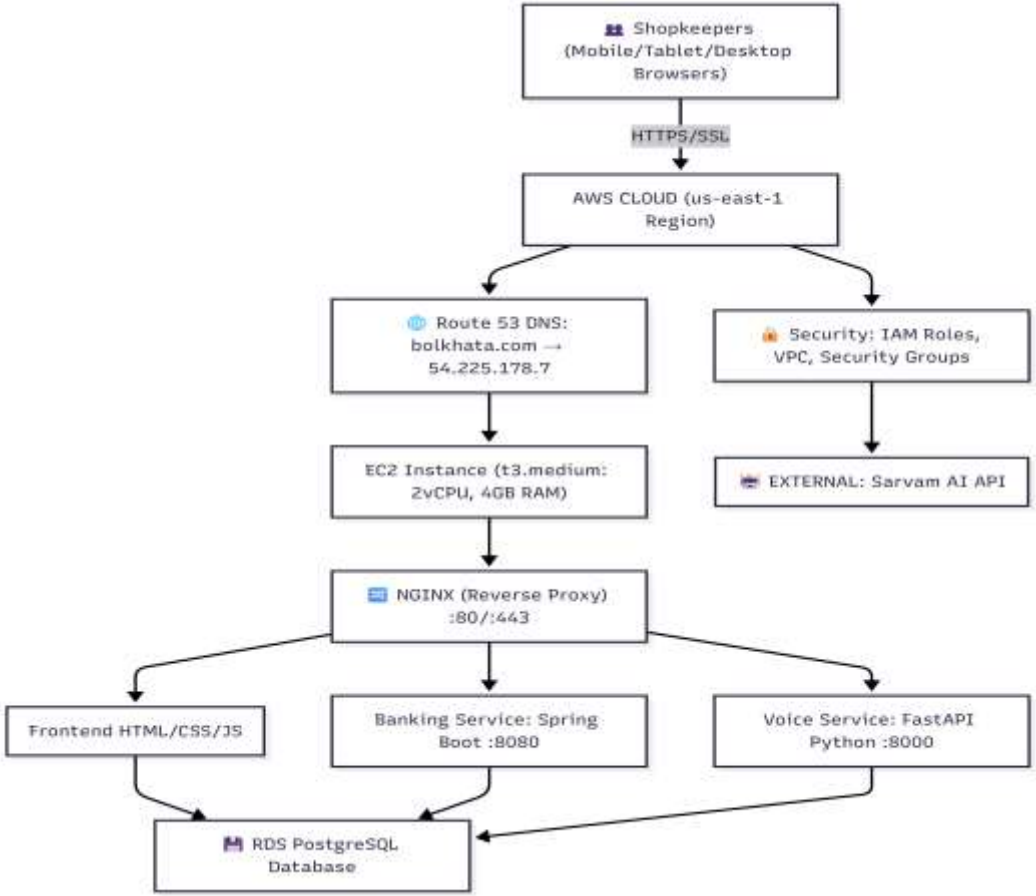
List of features offered by the solution

- Voice-First Transaction Entry - Record transactions by speaking naturally in Hindi, English, or Hinglish. Say "Ramesh ne 500 rupaye diye" and it's automatically logged in under 3 seconds.
- Intelligent Transaction Classification - Automatically distinguishes between three types: Cash Sales (SALE_PAID), Credit Sales (SALE_CREDIT), and Payment Received (PAYMENT_RECEIVED) based on conversational context.
- Customer Management - Automatic customer profile creation with running balance tracking, transaction history, and credit monitoring. No manual data entry required.
- Real-Time Dashboard - Live business analytics showing total sales, outstanding credits, payment collections, daily revenue trends, and customer-wise breakdowns.
- Digital Passbook Generation - Each customer gets an itemized digital passbook with complete transaction history that can be shared via WhatsApp for payment reminders and reconciliation.
- Multi-Language Support - Supports Hindi, English, and Hinglish with automatic translation, making it accessible across India's diverse linguistic landscape.
- Manual Entry Option - Traditional form-based entry available for users who prefer typing or need to add historical transactions.
- Audio Recording Storage - All voice transactions are recorded and stored with timestamps for audit trails and dispute resolution.
- Secure Authentication - User registration and login with encrypted password storage and session management.
- Cloud-Based Access - Accessible from any device via https://bolkhata.com with data synced in real-time to AWS RDS PostgreSQL.
- HTTPS Security - SSL encryption for all data transmission ensuring financial data privacy and security.

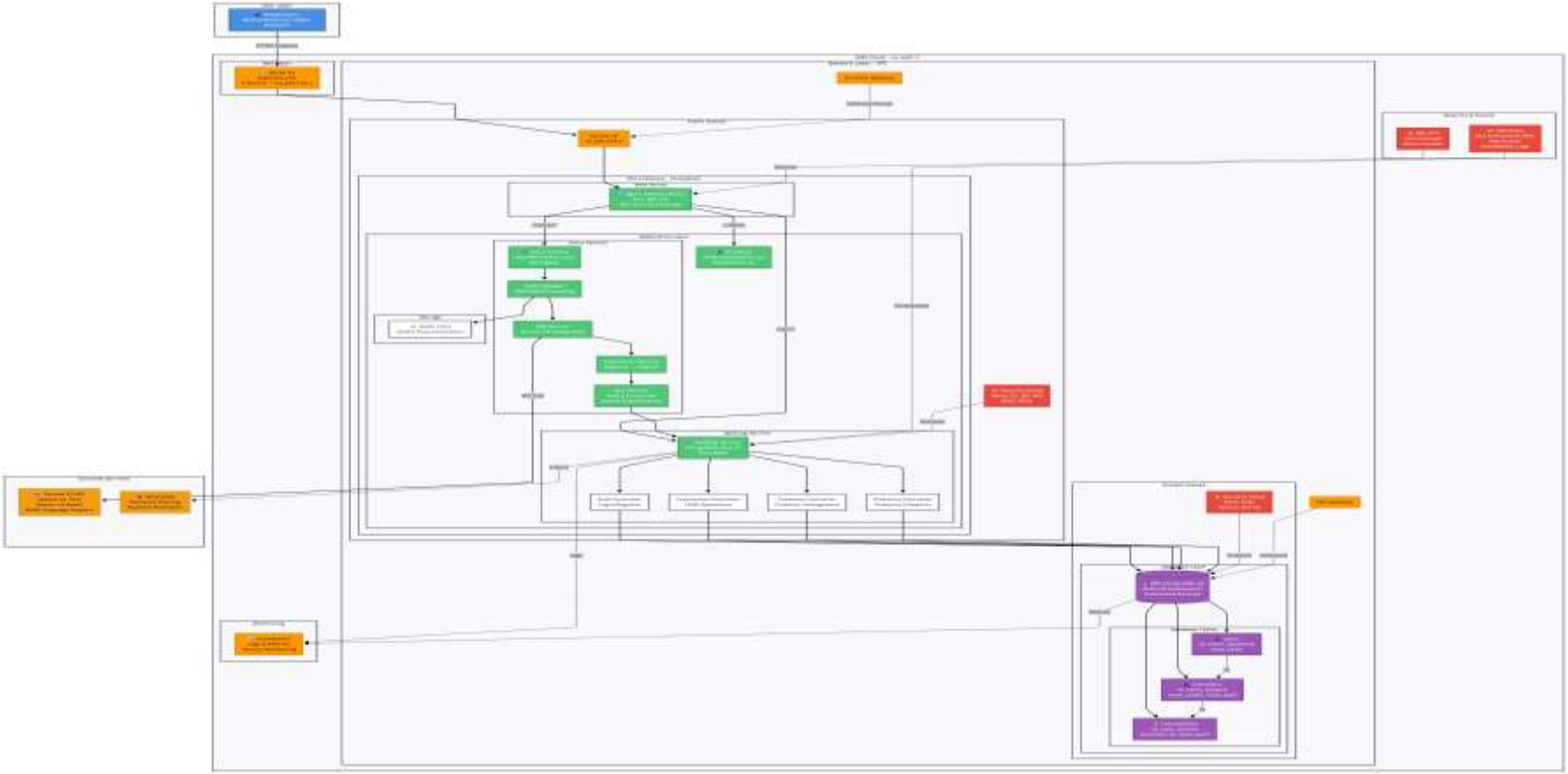
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:



Technologies utilized in the solution:

FRONTEND TECHNOLOGIES

HTML5, CSS3, JavaScript (ES6+) • Responsive Web Design (Mobile-first approach) • Web Audio API (for microphone access and audio recording) Fetch API (for REST API communication) • LocalStorage (for session management)

BACKEND TECHNOLOGIES

Banking Service: • Java 17 (Amazon Corretto) • Spring Boot 3.x (Framework) • Spring Data JPA (ORM for database operations) • Spring Security (Authentication & Authorization) • Maven (Build & Dependency Management) • Hibernate (JPA Implementation) • BCrypt (Password encryption)

Voice Service: • Python 3.11 • FastAPI (Web framework) • Uvicorn (ASGI server) • Pydantic (Data validation) • HTTPX (Async HTTP client) • Python-multipart (File upload handling) • Regex (Pattern matching for NLU)

ARTIFICIAL INTELLIGENCE & ML • Sarvam AI API (Speech-to-Text engine) • Saaras v3 Model (Multilingual ASR) • Custom NLU Engine (Entity extraction using regex patterns) • Natural Language Processing (Intent classification) • Translation Service (Regional language to English)

DATABASE: PostgreSQL 15.x (Relational database) • pg_trgm Extension (Fuzzy text search) • JDBC (Java Database Connectivity) • SQL (Structured Query Language)

AWS CLOUD SERVICES • EC2 (Elastic Compute Cloud) - t3.medium instances • RDS (Relational Database Service) - PostgreSQL • Route 53 (DNS management) • VPC (Virtual Private Cloud) • Security Groups (Firewall rules) • IAM (Identity & Access Management) • Elastic IP (Static IP addressing)

WEB SERVER & PROXY • Nginx (Reverse proxy & web server) • Let's Encrypt (SSL/TLS certificates) • Certbot (Certificate automation)

DEVOPS & DEPLOYMENT • Systemd (Service management) • Bash Scripting (Automation) • Git (Version control) • GitHub (Code repository) • SSH (Secure remote access) • AWS CLI (Cloud management)

SECURITY TECHNOLOGIES • HTTPS/TLS 1.3 (Encrypted communication) • SSL Certificates (Let's Encrypt) • BCrypt Hashing (Password security) • JWT/Session Tokens (Authentication) • CORS (Cross-Origin Resource Sharing) • Security Groups (Network-level firewall) • IAM Roles (Access control)

AUDIO PROCESSING • WAV Format (Audio storage) • MP3 Support (Audio compression) • Base64 Encoding (Audio transmission) • Audio File Storage System

API & INTEGRATION • RESTful API Architecture • JSON (Data interchange format) • Multipart Form Data (File uploads) • HTTP/HTTPS Protocols • WebSocket (Future real-time features)

LANGUAGES SUPPORTED • Hindi (hi-IN) • English (en-IN) • Gujarati (gu-IN) • Tamil (ta-IN) • Bengali (bn-IN) • Marathi (mr-IN) • Telugu (te-IN) • Kannada (kn-IN) • Malayalam (ml-IN) • Punjabi (pa-IN)

DEVELOPMENT TOOLS • Maven (Java build tool) • pip (Python package manager) • VS Code / IntelliJ IDEA (IDEs) • Postman (API testing) • Git Bash (Command line)

MONITORING & LOGGING • CloudWatch (AWS monitoring) • Systemd Logs (Service logging) • Application Logs (Custom logging)

Estimated implementation cost (optional):

Bol-Khata is engineered for extreme cost-efficiency without compromising production standards. By "rightsizing" our infrastructure and leveraging the AWS Free Tier, we have reduced the annual operating cost from a standard $2,000 to just $63.60. This 95% reduction allows us to offer a "Free Forever" tier for micro-merchants while maintaining high availability at https://bolkhata.com .

1. Core Infrastructure Breakdown (Annual)

- Compute & Logic (EC2): We utilize a t3.micro instance (1GB RAM) to host our Java and Python microservices. This is covered by 750 free hours/month, resulting in a $0.00 annual cost.
- Data Persistence (RDS): Our PostgreSQL 15.x database runs on a db.t3.micro instance with 20GB of SSD storage. Both are fully covered under the AWS Free Tier for the first 12 months.
- Static Assets & Audio (S3): All voice recordings and PDF passbooks are stored in AWS S3. The first 5GB of storage and 20,000 GET requests per month are free.
- Security & Networking: We use Let's Encrypt for automated, professional SSL certificates at $0.00. Our only fixed infrastructure cost is $6.60/year for Route 53 DNS hosting and a static Elastic IP.

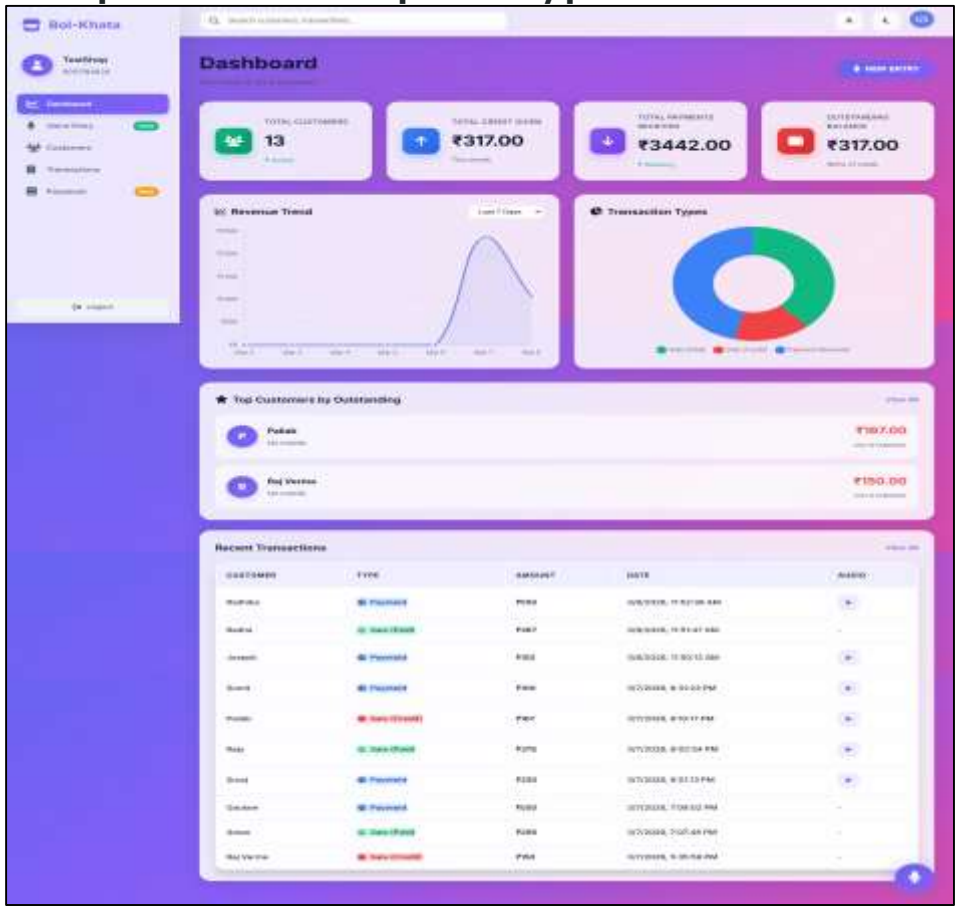
2. Operating Expenses & Unit Economics

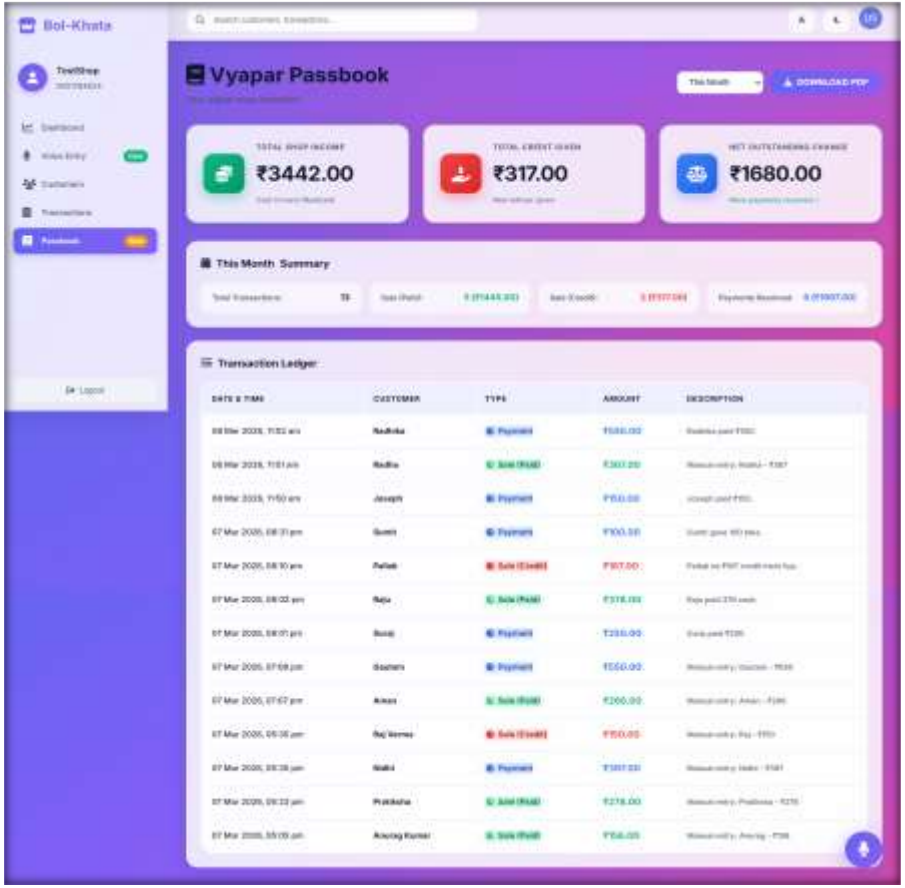
- Professional Identity: The bolkhata.com domain is a fixed investment of $12.00/year.
- AI Processing: We use a pay-per-use model for the Sarvam AI API, estimated at $45.00/year for a 100-user pilot.
- Development Investment: As a self-built project by TeamMAGNUS, our total development and maintenance cost remains $0.00.

3. Key Financial Metrics

- Total Year-1 Investment: $63.60 (approx. ₹5,300).
- Monthly Burn Rate: $5.30 (approx. ₹440)
- Cost per Active User: $0.63 / Year (at 100-user scale)

Snapshots of the prototype:





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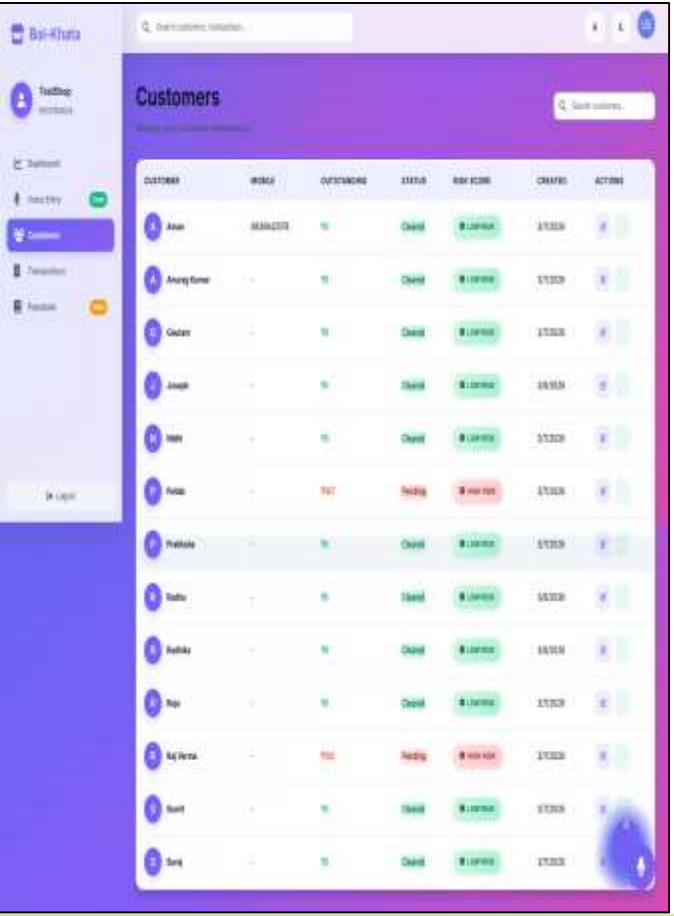
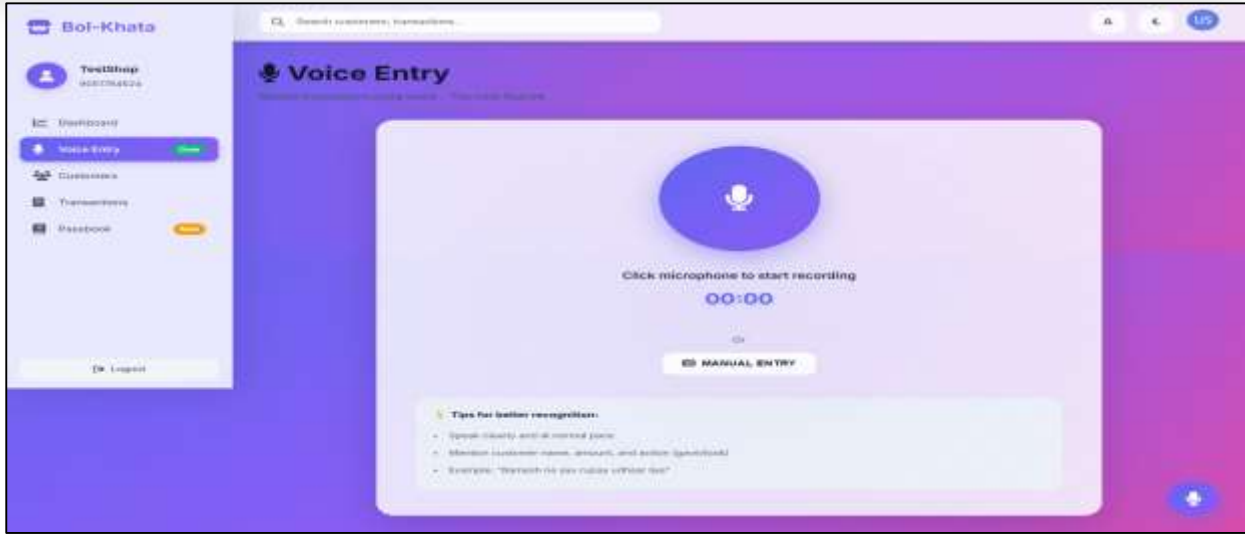
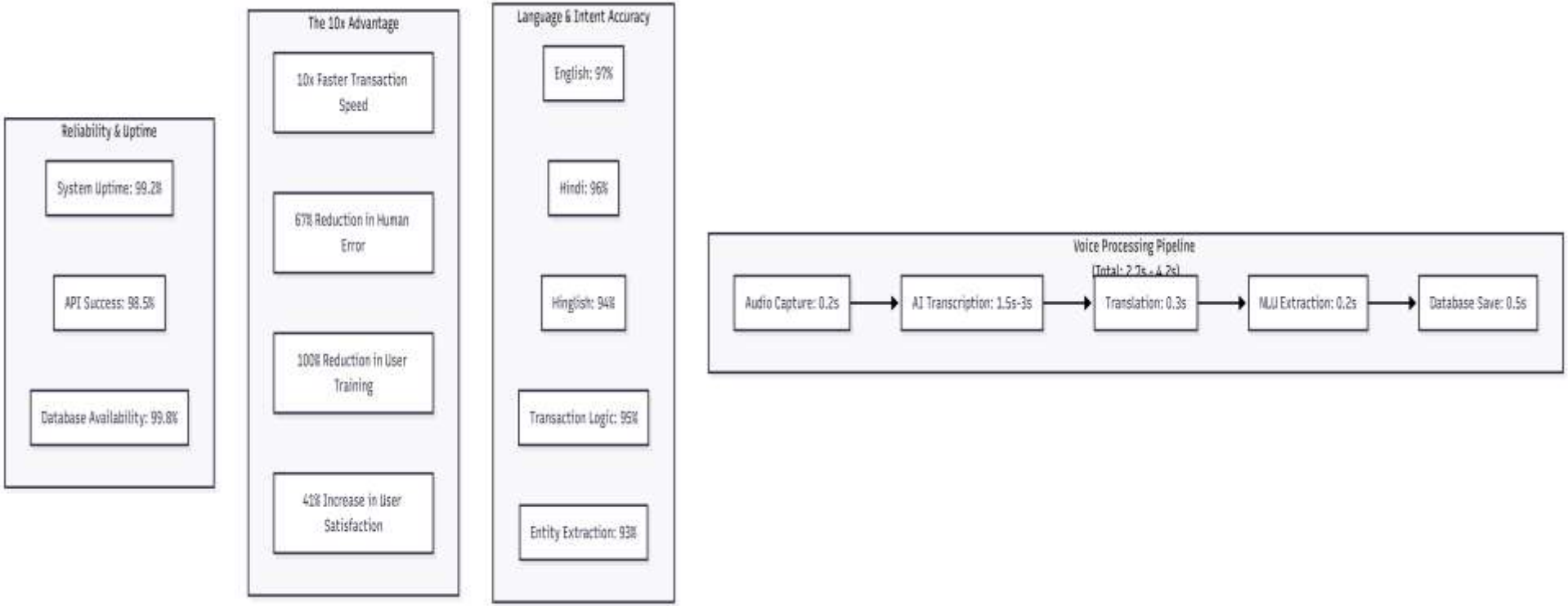




Prototype Performance report/Benchmarking:



Additional Details/Future Development (if any):

ENHANCED AI & NLU (Q2-Q3 2026)

Advanced Language Models • AWS Bedrock integration for conversational AI • Bhashini API for improved Indic language support • Custom LLM fine-tuning for financial domain • Multi-turn dialogue support for complex transactions • Context-aware entity extraction with 98%+ accuracy

Expanded Language Support • Add 5+ regional languages (Odia, Assamese, Kashmiri, Konkani, Sindhi) • Dialect recognition for better accuracy • Code-mixing support (Hinglish, Tanglish, etc.)

MOBILE & INTEGRATIONS (Q3-Q4 2026)

Mobile Applications • Native Android & iOS apps with offline mode • Biometric authentication • Push notifications for payment reminders

Third-Party Integrations • WhatsApp Business API for automated reminders • UPI payment integration (PhonePe, Google Pay, Paytm) • GST invoice generation • SMS gateway for notifications • Tally/QuickBooks sync

ADVANCED FEATURES (2027)

Business Intelligence • AI-powered cash flow forecasting • Predictive analytics for credit risk • Inventory management module • Multi-store/branch management • Automated reconciliation with bank statements

Collaboration & Security • Multi-user access with role-based permissions • Two-factor authentication (2FA) • Customer self-service portal • Blockchain-based audit trails

Prototype Assets:

https://bolkhata.com

- ➔ GitHub Public Repository Link

  https://github.com/Sumit648527/Bol_Khata

- ➔ Demo Video Link (Max: 3 Minutes)

  https://drive.google.com/file/d/1p1zupsnYgLj6dJN-8QYfK1k2xcCBFC0h/view?usp=sharing

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Thank You